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5 Module 5 | Examining Distributions (continued)

5.5 Module 5 | Case $Q \rightarrow Q$

Case $Q \rightarrow Q$: Two Quantitative Variables Here again is the role-type classification table for framing our discussion about the relationship between two variables:

		Response	
		Categorical	Quantitative
Explanatory	Categorical	✓ $C \rightarrow C$	✓ $C \rightarrow Q$
	Quantitative	$Q \rightarrow C$	$Q \rightarrow Q$

We are done with cases $C \rightarrow Q$ and $C \rightarrow C$, and now we will move on to case $Q \rightarrow Q$, where we examine the relationship between two quantitative variables.

5.5.5 Module 5 | Scatterplot (1 of 5)

Learning Objectives

- Graphically display the relationship between two quantitative variables and describe: a) the overall pattern, and b) striking deviations from the pattern.

In the previous two cases we had a categorical explanatory variable, and therefore exploring the relationship between the two variables was done by comparing the distribution of the response variable for each category of the explanatory variable:

- In case $C \rightarrow Q$ we compared distributions of the quantitative response.
- In case $C \rightarrow C$ we compared distributions of the categorical response.

Case $Q \rightarrow Q$ is different in the sense that both variables (in particular the explanatory variable) are quantitative, and therefore, as you'll discover, this case will require a different kind of treatment and tools. Let's start with an example:

 Example (A) Highway Signs

A Pennsylvania research firm conducted a study in which 30 drivers (of ages 18 to 82 years old) were sampled, and for each one, the maximum distance (in feet) at which he/she could read a newly designed sign was determined. The goal of this study was to explore the relationship between a driver's *age* and the *maximum distance* at which signs were legible, and then use the study's findings to improve safety for older drivers. (Reference: Utts and Heckard, *Mind on Statistics* (2002). Originally source: Data collected by Last Resource, Inc, Bellfonte, PA.)

Since the purpose of this study is to explore the effect of age on maximum legibility distance,

- the *explanatory* variable is *Age*, and
- the *response* variable is *Distance*.

Here is what the raw data look like:

Explanatory

Response

	Age	Distance
Driver 1	18	510
Driver 2	32	410
Driver 3	55	420
Driver 4	23	510
.	.	.
.	.	.
.	.	.
Driver 30	82	360

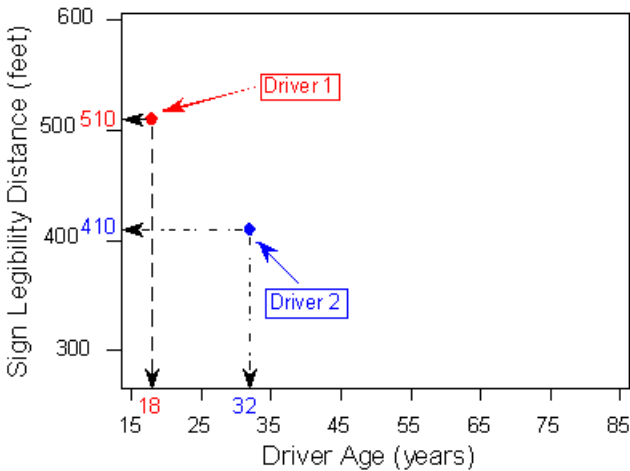
Note that the data structure is such that for each individual (in this case driver 1....driver 30) we have a pair of values (in this case representing the driver's age and distance). We can therefore think about these data as 30 pairs of values: (18, 510), (32, 410), (55, 420), ... , (82, 360).

The first step in exploring the relationship between driver age and sign legibility distance is to create an appropriate and informative graphical display. The appropriate graphical display for examining the relationship between two quantitative variables is the *scatterplot*. Here is how a scatterplot is constructed for our example:

To create a scatterplot, each pair of values is plotted, so that the value of the explanatory variable (X) is plotted on the horizontal axis, and the value of the response variable (Y) is plotted on the vertical axis. In other words, each individual (driver, in our example) appears on the scatterplot as a single point whose X-coordinate is the value of

the explanatory variable for that individual, and whose Y-coordinate is the value of the response variable. Here is an illustration:

	Age (X)	Distance (Y)
Driver 1	18	510
Driver 2	32	410
Driver 3	55	420
Driver 4	23	510
.	.	.
.	.	.
.	.	.
Driver 30	82	360



And here is the completed scatterplot:



Comment It is important to mention again that when creating a scatterplot, the explanatory variable should always be plotted on the horizontal X-axis, and the response variable should be plotted on the vertical Y-axis. If in a specific example we do not have a clear distinction between explanatory and response variables, each of the variables can be plotted on either axis.

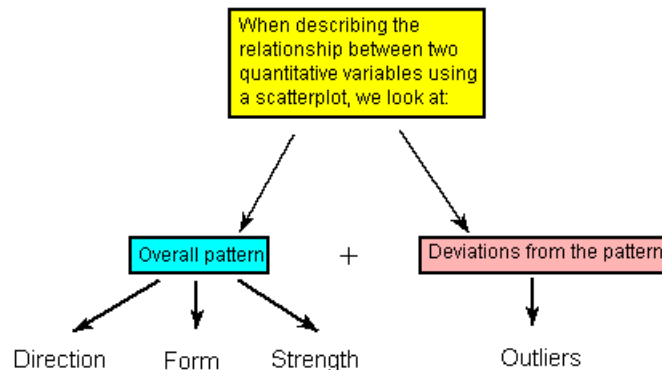
5.5.6 Module 5 | Scatterplot (2 of 5)

Learning Objectives

- Graphically display the relationship between two quantitative variables and describe: a) the overall pattern, and b) striking deviations from the pattern.

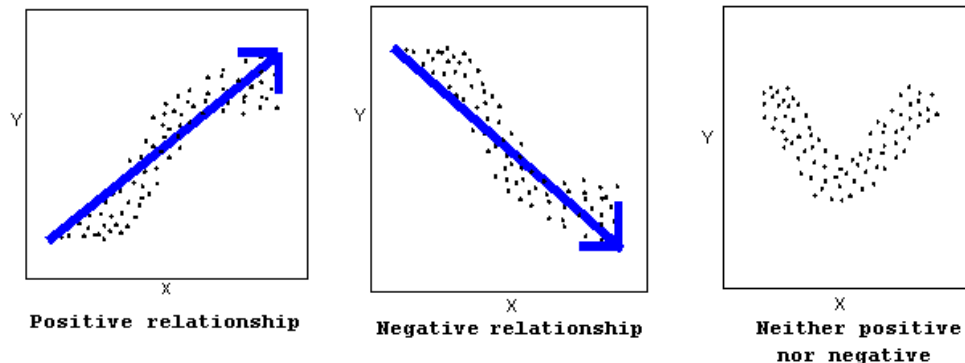
Interpreting the scatterplot How do we explore the relationship between two quantitative variables using the scatterplot? What should we look at, or pay attention to?

Recall that when we described the distribution of a single quantitative variable with a histogram, we described the overall pattern of the distribution (shape, center, spread) and any deviations from that pattern (outliers). *We do the same thing with the scatterplot.* The following figure summarizes this point:



As the figure explains, when describing the *overall pattern* of the relationship we look at its direction, form and strength.

- The *direction* of the relationship can be positive, negative, or neither:

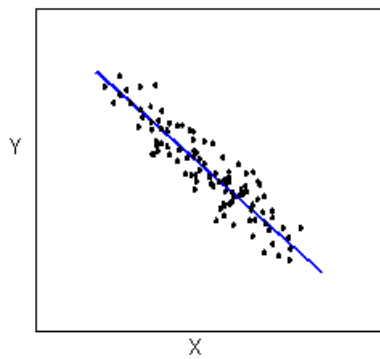


A *positive (or increasing) relationship* means that an increase in one of the variables is associated with an increase in the other.

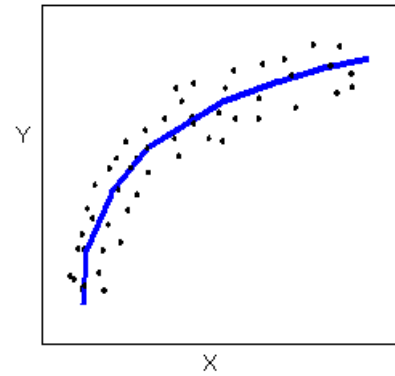
A *negative (or decreasing) relationship* means that an increase in one of the variables is associated with a decrease in the other.

Not all relationships can be classified as either positive or negative.

- The *form* of the relationship is its general shape. When identifying the form, we try to find the simplest way to describe the shape of the scatterplot. There are many possible forms. Here are a couple that are quite common:

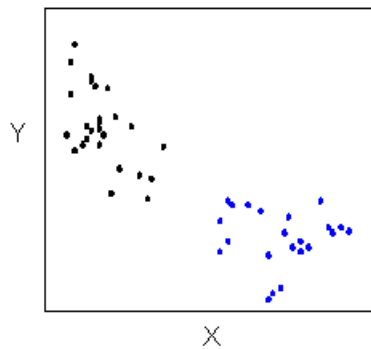


Relationships with a *linear* form are most simply described as points scattered about a line.

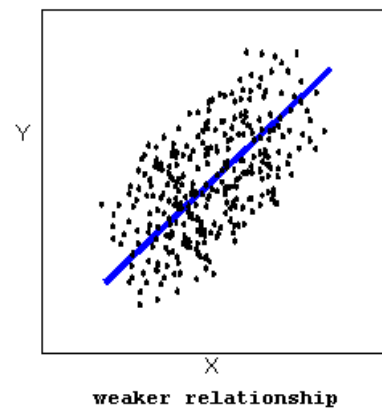
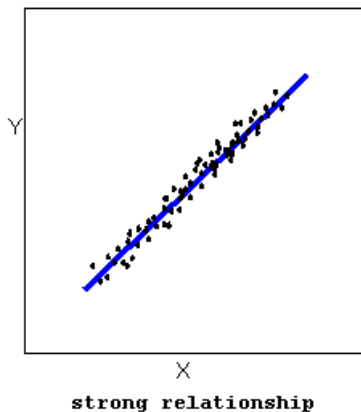


Relationships with a *curvilinear* form are most simply described as points dispersed around the same curved line.

There are many other possible forms for the relationship between two quantitative variables, but linear and curvilinear forms are quite common and easy to identify. Another form-related pattern that we should be aware of is clusters in the data:

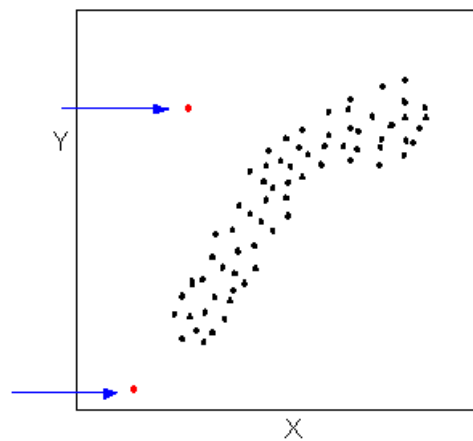


- The *strength* of the relationship is determined by how closely the data follow the form of the relationship. Let's look, for example, at the following two scatterplots displaying positive, linear relationships:



The strength of the relationship is determined by how closely the data points follow the form. We can see that in the top scatterplot the data points follow the linear pattern quite closely. This is an example of a strong relationship. In the bottom scatterplot, the points also follow the linear pattern, but much less closely, and therefore we can say that the relationship is weaker. In general, though, assessing the strength of a relationship just by looking at the scatterplot is quite problematic, and we need a numerical measure to help us with that. We will discuss that later in this section.

Data points that *deviate from the pattern* of the relationship are called *outliers*. We will see several examples of outliers during this section. Two outliers are illustrated in the scatterplot below:



5.5.7 Module 5 | Scatterplot (3 of 5)

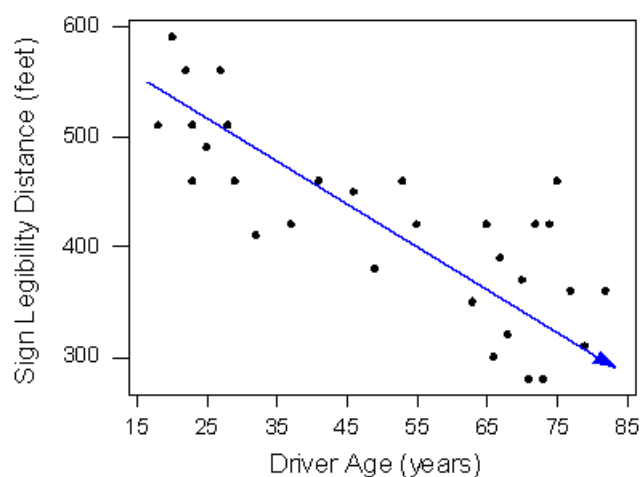
Learning Objectives

- Graphically display the relationship between two quantitative variables and describe: a) the overall pattern, and b) striking deviations from the pattern.

Let's go back now to our example, and use the scatterplot to examine the relationship between the age of the driver and the maximum sign legibility distance. Here is the scatterplot:



The direction of the relationship is *negative*, which makes sense in context, since as you get older your eyesight weakens, and in particular older drivers tend to be able to read signs only at lesser distances. An arrow drawn over the scatterplot illustrates the negative direction of this relationship:



The form of the relationship seems to be *linear*. Notice how the points tend to be scattered about the line. Although, as we mentioned earlier, it is problematic to assess the strength without a numerical measure, the relationship appears to be *moderately strong*, as the data is fairly tightly scattered about the line. Finally, all the data points seem to “obey” the pattern – there *do not appear to be any outliers*.

5.5.8 Module 5 | Scatterplot (4 of 5)

Learning Objectives

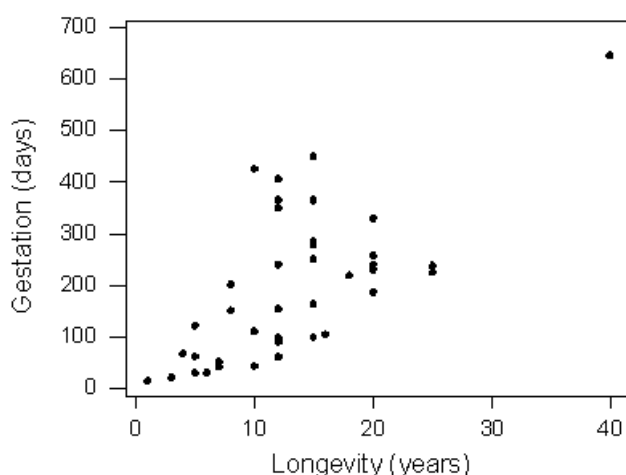
- Graphically display the relationship between two quantitative variables and describe: a) the overall pattern, and b) striking deviations from the pattern.

We will now look at two more examples:

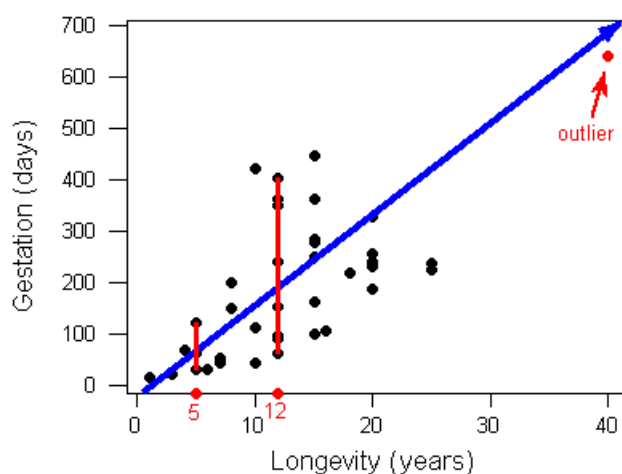
🔗 Example (B) Average Gestation Period

The average gestation period, or time of pregnancy, of an animal is closely related to its longevity (the length of its lifespan.) Data on the average gestation period and longevity (in captivity) of 40 different species of animals have been examined, with the purpose of examining how the gestation period of an animal is related to (or can be predicted from) its longevity. (Source: Rossman and Chance. (2001). Workshop statistics: Discovery with data and Minitab. Original source: The 1993 world almanac and book of facts).

Here is the scatterplot of the data.



What can we learn about the relationship from the scatterplot? The direction of the relationship is *positive*, which means that animals with longer life spans tend to have longer times of pregnancy (this makes intuitive sense). An arrow drawn over the scatterplot below illustrates this:

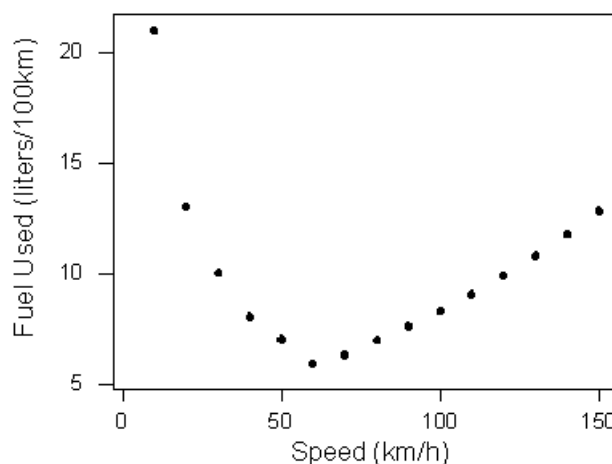


The form of the relationship is again essentially *linear*. There appears to be *one outlier*, indicating an animal with an exceptionally long longevity and gestation period. (This animal happens to be the elephant.) Note that while this outlier definitely deviates from the rest of the data in term of its magnitude, it *does* follow the direction of the data.

Comment Another feature of the scatterplot that is worth observing is how the variation in gestation increases as longevity increases. This fact is illustrated by the two red vertical lines at the bottom left part of the graph. Note that the gestation periods for animals who live 5 years range from about 30 days up to about 120 days. On the other hand, the gestation period of animals who live 12 years varies much more, and ranges from about 60 days up to more than 400 days.

Example (C) Fuel Usage

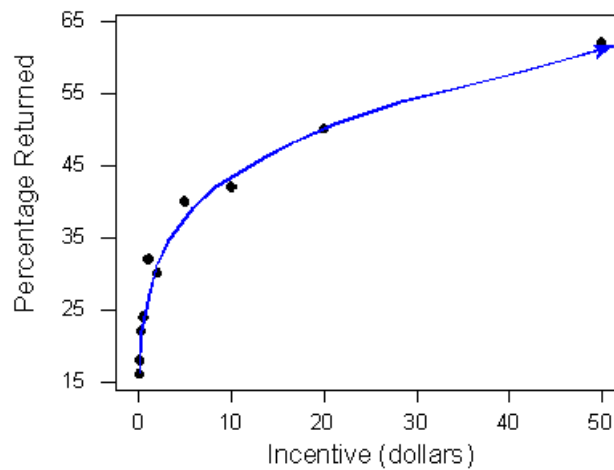
As a third example, consider the relationship between the average amount of fuel used (in liters) to drive a fixed distance in a car (100 kilometers), and the speed at which the car is driven (in kilometers per hour). (Source: Moore and McCabe, (2003). Introduction to the practice of statistics. Original source: T.N. Lam. (1985). “Estimating fuel consumption for engine size,” Journal of Transportation Engineering, vol. 111)



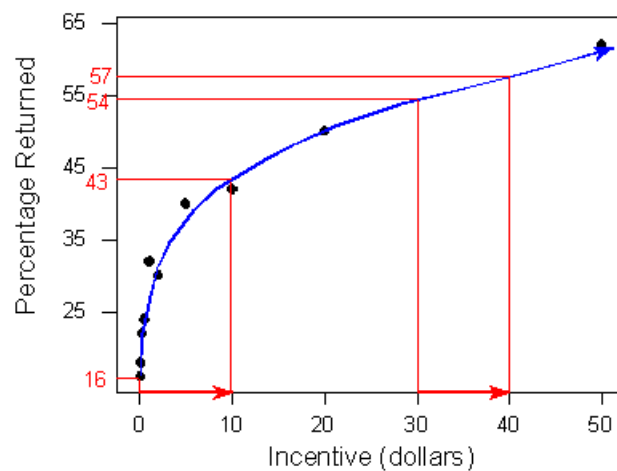
The data describe a relationship that decreases and then increases – the amount of fuel consumed decreases rapidly to a minimum for a car driving 60 kilometers per hour, and then increases gradually for speeds exceeding 60 kilometers per hour. This suggests that the speed at which a car economizes on fuel the most is about 60 km/h. This forms a curvilinear relationship that seems to be very strong, as the observations seem to perfectly fit the curve. Finally, there do not appear to be any outliers.

Example (D)

Background A study examined how the percentage of participants who completed a survey is related to the monetary incentive that researchers promised to participants. Consider the relationship between these two quantitative variables, displayed in the scatterplot below.



The positive relationship definitely makes sense in context, but what is the interpretation of the curvilinear form in the context of the problem? How can we explain (in context) the fact that the relationship seems at first to be increasing very rapidly, but then slows down? The following graph will help us:



Note that when the monetary incentive increases from \$0 to \$10, the percentage of returned surveys increases sharply – an increase of 27% (from 16% to 43%). However, the same increase of \$10 from \$30 to \$40 doesn’t result in the same dramatic increase in the percentage of returned surveys – it results in an increase of only 3% (from 54% to 57%). The form displays the phenomenon of “diminishing returns” – a return rate that after a certain point fails to increase proportionately to additional outlays of investment. \$10 is worth more to people relative to \$0 than it is relative to \$30.

5.5.9 Module 5 | Scatterplot (5 of 5)

Learning Objectives

- Graphically display the relationship between two quantitative variables and describe: a) the overall pattern, and b) striking deviations from the pattern.

A Labeled Scatterplot In certain circumstances, it may be reasonable to indicate different subgroups or categories within the data on the scatterplot, by labeling each subgroup differently. The result is called a *labeled scatterplot*, and can provide further insight about the relationship we are exploring. here is an example.

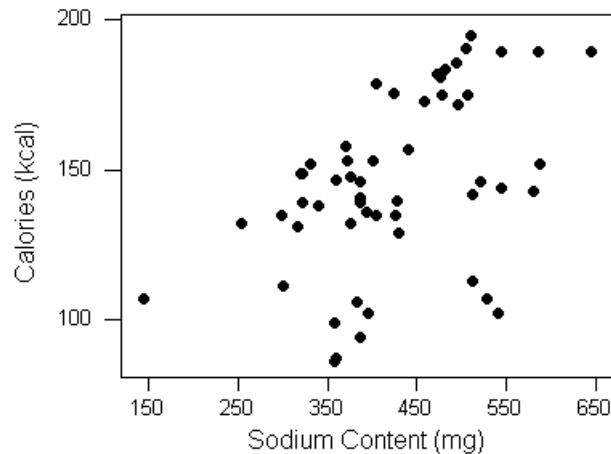
Example (E) Hot Dogs

Recall the hot dog example from case C→Q, in which 54 major hot dog brands were examined. In this study, both the *calorie content* and the *sodium level* of each brand was recorded, as well as the *type* of hot dog: beef, poultry, and meat (mostly pork and beef, but up to 15% poultry meat). In this example we will explore the

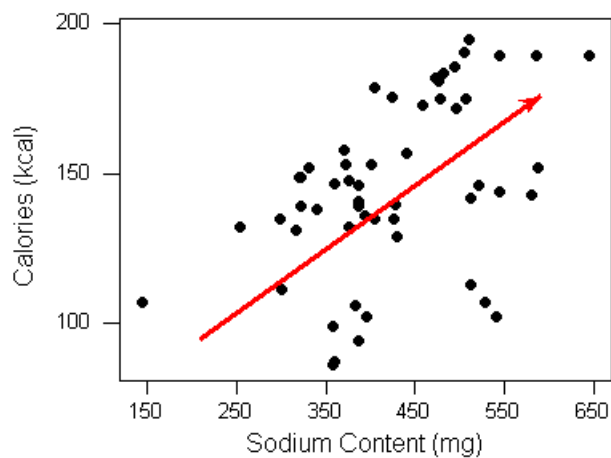
relationship between the sodium level and calorie content of hot dogs, and we will label the three different types of hot dogs to create a labeled scatterplot.

5.5.10 Creating a Labeled Scatterplot

The scatterplot below displays the relationship between the sodium and calorie content of 54 brands of hot dogs. Note that in this example there is no clear explanatory-response distinction, and we decided to have sodium content as the explanatory variable, and calorie content as the response variable.

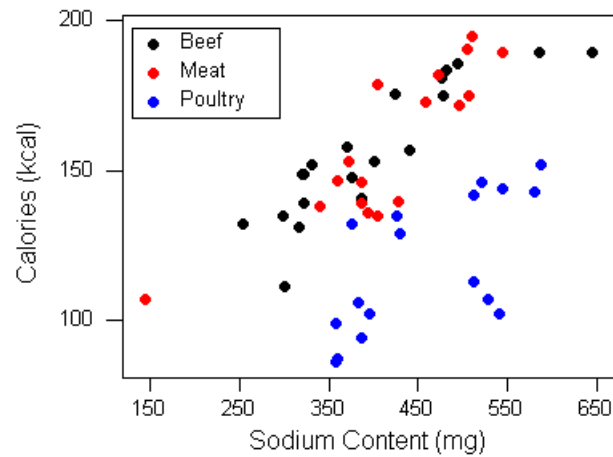


The scatterplot displays a positive relationship, which means that hot dogs containing more sodium tend to be higher in calories.

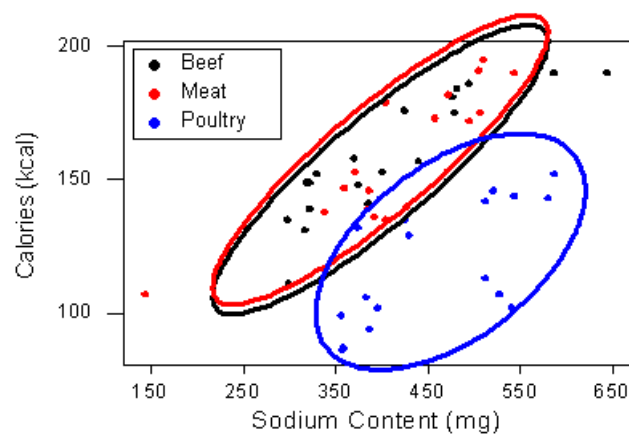


The form of the relationship, however, is kind of hard to determine. Maybe if we label the scatterplot, indicating the type of hot dogs, we will get a better understanding of the form.

Here is the labeled scatterplot, with the three different colors representing the three types of hot dogs, as indicated.

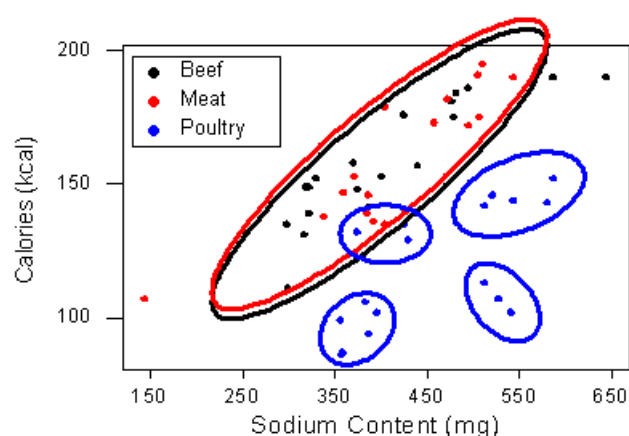


The display does give us more insight about the form of the relationship between sodium and calorie content.



It appears that there is a positive relationship within all three types. In other words, we can generally expect hot dogs that are higher in sodium to be higher in calories, no matter what type of hot dog we consider. In addition, we can see that hot dogs made of poultry (indicated in blue) are generally lower in calories. This is a result we have seen before.

Interestingly, it appears that the form of the relationship specifically for poultry is further clustered, and



we can only speculate about whether there is another categorical variable that describes these apparent sub-categories of poultry hot dogs.

5.5.11 Let's Summarize

- The relationship between two quantitative variables is visually displayed using the *scatterplot*, where each point represents an individual. We always plot the explanatory variable on the horizontal X axis, and the response variable on the vertical Y axis.
- When we explore a relationship using the scatterplot we should describe the *overall pattern* of the relationship and any *deviations* from that pattern. To describe the overall pattern consider the *direction*, *form* and *strength* of the relationship. Assessing the strength just by looking at the scatterplot can be problematic; using a numerical measure to determine strength will be discussed later in this course.
- Adding labels to the scatterplot that indicate different groups or categories within the data might help us get more insight about the relationship we are exploring.

5.5.12 Module 5 | Linear Relationships (1 of 8)

Learning Objectives

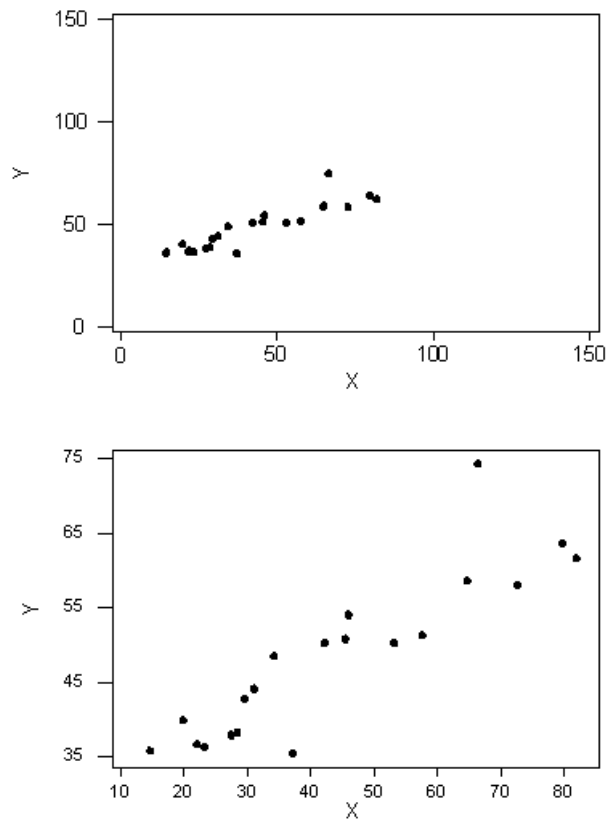
- Interpret the value of the correlation coefficient, and be aware of its limitations as a numerical measure of the association between two quantitative variables.

Introduction So far we have visualized relationships between two quantitative variables using scatterplots, and described the overall pattern of a relationship by considering its direction, form, and strength. We noted that assessing the strength of a relationship just by looking at the scatterplot is quite difficult, and therefore we need to supplement the scatterplot with some kind of numerical measure that will help us assess the strength.

In this part, we will restrict our attention to the *special case of relationships that have a linear form*, since they are quite common and relatively simple to detect. More importantly, there exists a numerical measure that assesses the strength of the linear relationship between two quantitative variables with which we can supplement the scatterplot. We will introduce this numerical measure here and discuss it in detail.

Even though from this point on we are going to focus only on linear relationships, it is important to remember that *not every relationship between two quantitative variables has a linear form*. We have actually seen several examples of relationships that are not linear. The statistical tools that will be introduced here are *appropriate only for examining linear relationships*, and as we will see, when they are used in nonlinear situations, these tools can lead to errors in reasoning.

Let's start with a motivating example. Consider the following two scatterplots.



We can see that in both cases, the direction of the relationship is *positive* and the form of the relationship is *linear*. What about the strength? Recall that the strength of a relationship is the extent to which the data follow its form.

Assessing the strength of the linear relationship from a scatterplot alone is problematic, since our judgment might be affected by the scale on which the values are plotted. This provides a motivation for the *need* to supplement the scatterplot with a *numerical measure* that will *measure the strength* of the linear relationship between two quantitative variables.

5.5.13 Module 5 | Linear Relationships (2 of 8)

Learning Objectives

- Interpret the value of the correlation coefficient, and be aware of its limitations as a numerical measure of the association between two quantitative variables.

The Correlation Coefficient – r The numerical measure that assesses the strength of a linear relationship is called the *correlation coefficient* and is denoted by r . We will

- Define the correlation r .
- Discuss the calculation of r .
- Explain how to interpret the value of r .
- Talk about some of the properties of r .

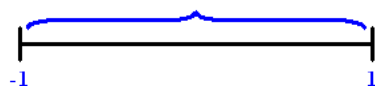
Definition: The correlation coefficient (r) is a numerical measure that measures the *strength* and *direction* of a linear relationship between two quantitative variables.

Calculation: r is calculated using the following formula:
$$r = \frac{1}{n-1} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{S_x} \right) \left(\frac{y_i - \bar{y}}{S_y} \right).$$

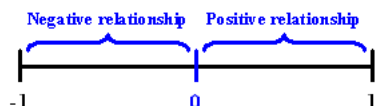
However, the calculation of the correlation (r) is not the focus of this course. We will use a statistics package to calculate r for us, and the emphasis of this course is on the *interpretation* of its value.

Interpretation Once we obtain the value of r , its interpretation with respect to the strength of linear relationships is quite simple, as this walkthrough illustrates¹:

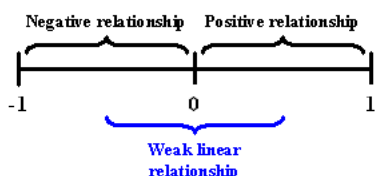
The value of r ranges from -1 to 1.



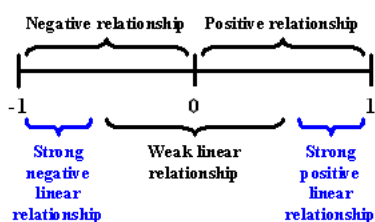
Negative values of r indicate a negative direction for a linear relationship, and positive values of r indicate a positive direction for a linear relationship.



Values of r that are close to 0—either negative or positive—indicate a weak linear relationship.



Values that are close to -1 or close to 1 indicate a strong linear relationship, either negative or positive.



5.5.14 Module 5 | Linear Relationships (3 of 8)

Learning Objectives

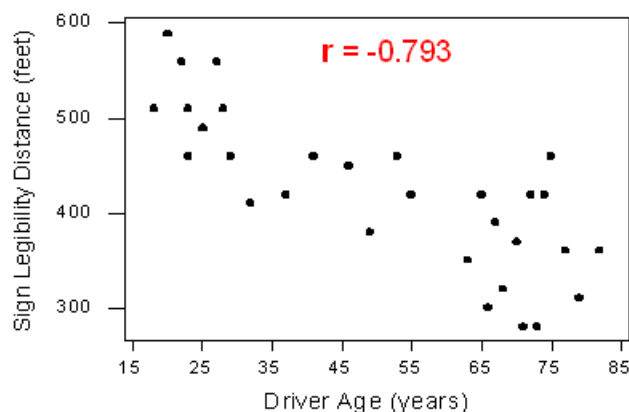
- Interpret the value of the correlation coefficient, and be aware of its limitations as a numerical measure of the association between two quantitative variables.

Now that we understand the use of r as a numerical measure for assessing the direction and strength of linear relationships between quantitative variables, we will look at a few examples.

¹This may be viewed as a video at <https://youtu.be/Bt-Ey2ebfvs>

Example (F) Highway Sign Visibility

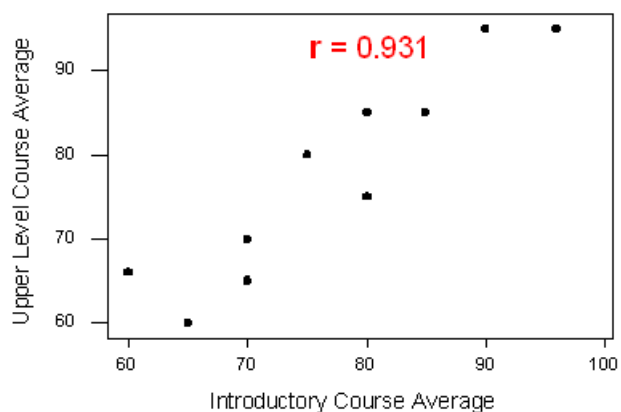
Earlier, we used the scatterplot below to find a *negative linear* relationship between the age of a driver and the maximum distance at which a highway sign was legible. What about the strength of the relationship? It turns out that the correlation between the two variables is $r = -0.793$.



Since $r < 0$, it confirms that the direction of the relationship is negative (although we really didn't need r to tell us that). Since r is relatively close to -1 , it suggests that the relationship is moderately strong. In context, the negative correlation confirms that the maximum distance at which a sign is legible generally decreases with age. Since the value of r indicates that the linear relationship is moderately strong, but not perfect, we can expect the maximum distance to vary somewhat, even among drivers of the same age.

Example (G) Statistics Courses

A statistics department is interested in tracking the progress of its students from entry until graduation. As part of the study, the department tabulates the performance of 10 students in an introductory course and in an upper-level course required for graduation. What is the relationship between the students' course averages in the two courses? Here is the scatterplot for the data:



The scatterplot suggests a relationship that is *positive* in direction, *linear* in form, and seems quite strong. The value of the correlation that we find between the two variables is $r = 0.931$, which is very close to 1, and thus confirms that indeed the linear relationship is very strong.

Comment Note that in both examples we supplemented the scatterplot with the correlation (r). Now that we have the correlation (r), why do we still need to look at a scatterplot when examining the relationship between two quantitative variables?

The *correlation* coefficient can *only* be interpreted as the *measure of the strength of a linear relationship*, so we need the scatterplot to verify that the relationship indeed looks linear. This point and its importance will be clearer after we examine a few properties of r .

5.5.15 Module 5 | Linear Relationships (4 of 8)

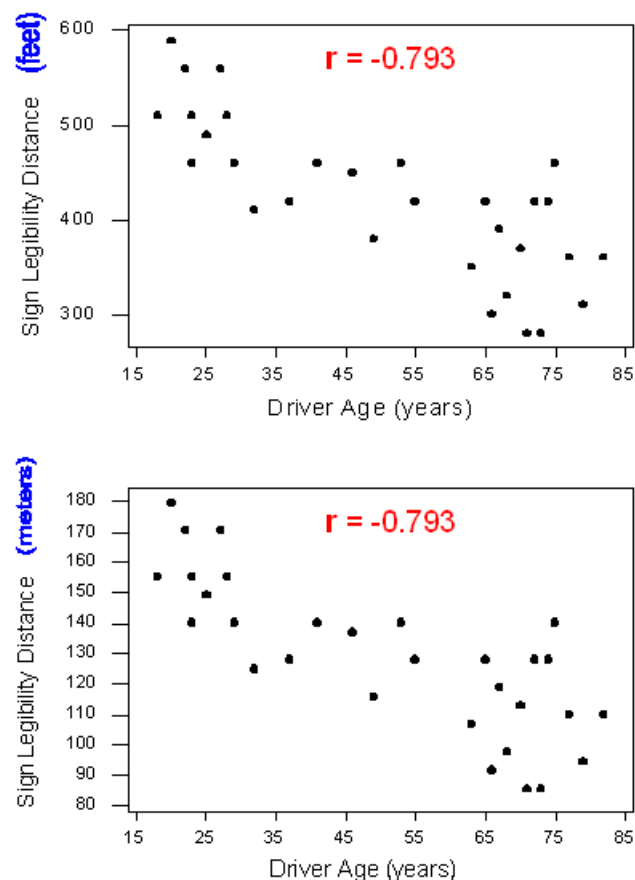
Learning Objectives

- Interpret the value of the correlation coefficient, and be aware of its limitations as a numerical measure of the association between two quantitative variables.

Properties of r We now discuss and illustrate several important properties of the correlation coefficient as a numerical measure of the strength of a linear relationship.

- The correlation does not change when the units of measurement of either one of the variables change. In other words, if we change the units of measurement of the explanatory variable and/or the response variable, the change has *no effect on the correlation (r)*.

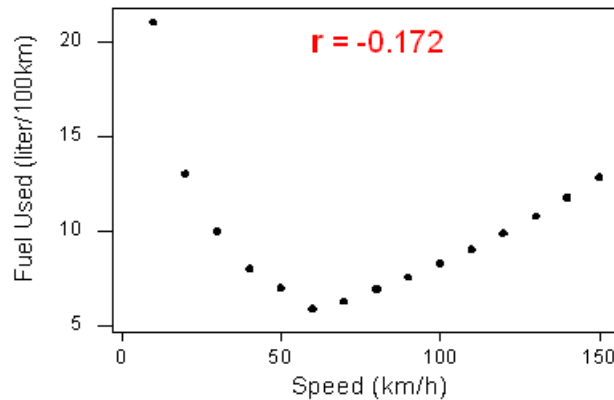
To illustrate, following are two versions of the scatterplot of the relationship between sign legibility distance and driver's age:



The top scatterplot displays the original data where the maximum distances is measured in *feet*. The bottom scatterplot displays the same relationship but with maximum distances changed to *meters*. Notice that the Y-values have changed, but the correlations are the same. This example illustrates how changing the units of measurement of the response variable has no effect on r , but as we indicated above, the same is true for changing the units of the explanatory variable, or of both variables.

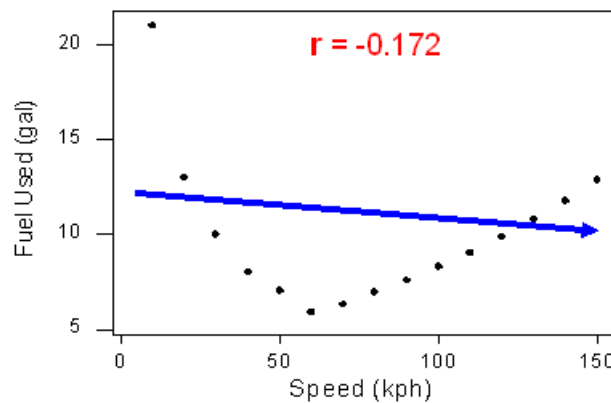
This might be a good place to comment that the correlation (r) is *unitless*. It is just a number.

- The correlation measures only the *strength* of a linear relationship between two variables. It *ignores* any other type of relationship, no matter how strong it is. For example, consider the relationship between the average fuel usage of driving a fixed distance in a car, and the speed at which the car drives:



Our data describe a fairly simple curvilinear relationship: the amount of fuel consumed decreases rapidly to a minimum for a car driving 60 kilometers per hour, and then increases gradually for speeds exceeding 60 kilometers per hour. The relationship is very strong, as the observations seem to perfectly fit the curve.

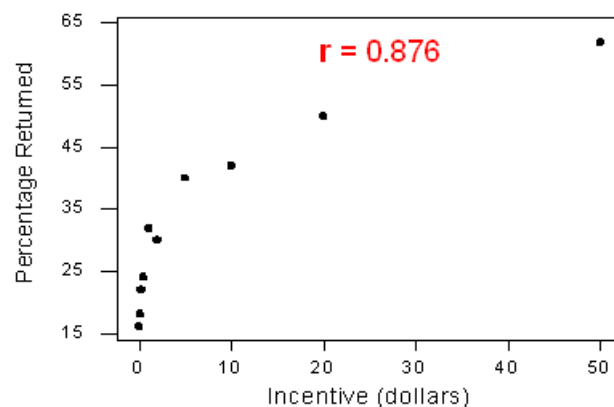
Although the relationship is strong, the correlation $r = -0.172$ indicates a weak *linear* relationship. This makes sense considering that the data fails to adhere closely to a linear form:



The correlation is useless for assessing the strength of any type of relationship that is not linear (including relationships that are curvilinear, such as the one in our example). Beware, then, of interpreting the fact that r is close to 0 as an indicator of a weak relationship rather than a weak *linear* relationship. This example also illustrates how important it is to *always look at the data in the scatterplot* because, as in our example, there might be a strong nonlinear relationship that r does not indicate.

Since the correlation was nearly zero when the form of the relationship was not linear, we might ask if the correlation can be used to determine whether or not a relationship is linear.

3. The correlation by itself is *not* sufficient to determine whether a relationship is linear. To see this, let's consider the study that examined the effect of monetary incentives on the return rate of questionnaires. Below is the scatterplot relating the percentage of participants who completed a survey to the monetary incentive that researchers promised to participants, in which we find a *strong curvilinear relationship*:



The relationship is curvilinear, yet the correlation $r = 0.876$ is quite close to 1.

In the last two examples, we have seen two very strong curvilinear relationships, one with a correlation close to 0 and one with a correlation close to 1. Therefore, the correlation alone does not indicate whether a relationship is linear. The important principle here is:

Always look at the data!

4. The correlation is heavily influenced by outliers. The way in which the outlier influences the correlation depends on whether the outlier is consistent with the pattern of the linear relationship.

5.5.16 Module 5 | Linear Relationships (5 of 8)

Learning Objectives

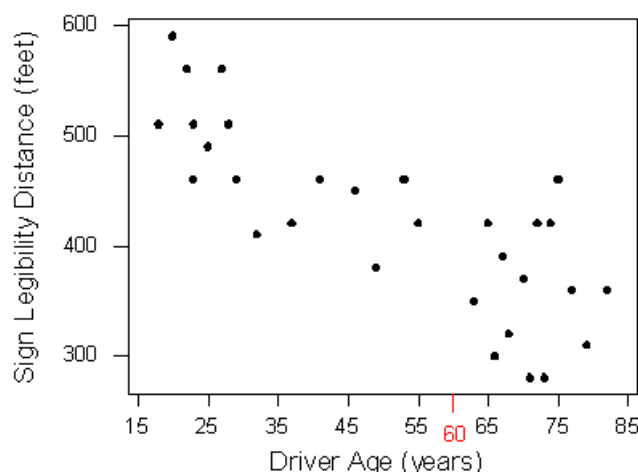
- In the special case of linear relationship, use the least squares regression line as a summary of the overall pattern, and use it to make predictions.

Linear Regression: Summarizing the Pattern of the Data with a Line So far we've used the scatterplot to describe the relationship between two quantitative variables, and in the special case of a linear relationship, we have supplemented the scatterplot with the correlation (r). The correlation, however, doesn't fully characterize the linear relationship between two quantitative variables – it only measures the strength and direction. We often want to describe more precisely how one variable changes with the other (by “more precisely,” we mean more than just the direction), or *predict* the value of the response variable for a given value of the explanatory variable. In order to be able to do that, we need to summarize the linear relationship with a line that best fits the linear pattern of the data. In the remainder of this section, we will introduce a way to find such a line, learn how to interpret it, and use it (cautiously) to make predictions.

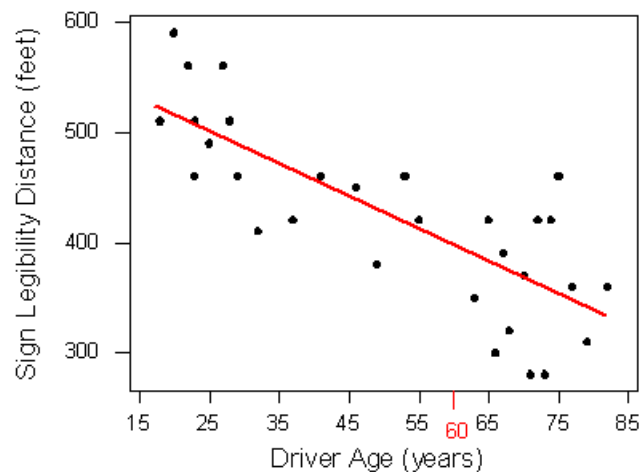
Again, let's start with a motivating example:

Earlier, we examined the linear relationship between the age of a driver and the maximum distance at which a highway sign was legible, using both a scatterplot and the correlation coefficient. Suppose a government agency wanted to predict the maximum distance at which the sign would be legible for 60-year-old drivers, and thus make sure that the sign could be used safely and effectively.

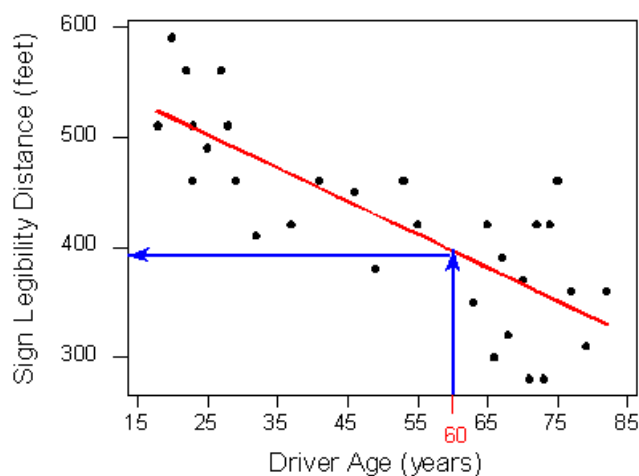
How would we make this prediction? Here again is the scatterplot of driver ages and maximum distances at which a sign is legible. The age for which an agency wishes to predict the legibility distance, 60, is marked in red. How can we make this prediction?



It would be useful if we could find a line (such as the one that is presented on the scatterplot) that represents the general pattern of the data, because then,



we would simply use this line to find the distance that corresponds to an age of 60 like this:



and predict that 60-year-old drivers could see the sign from a distance of just under 400 feet.

How and why did we pick this particular line (the one shown in red in the above walkthrough) to describe the dependence of the maximum distance at which a sign is legible upon the age of a driver? What line exactly did we choose? We will return to this example once we can answer that question with a bit more precision.

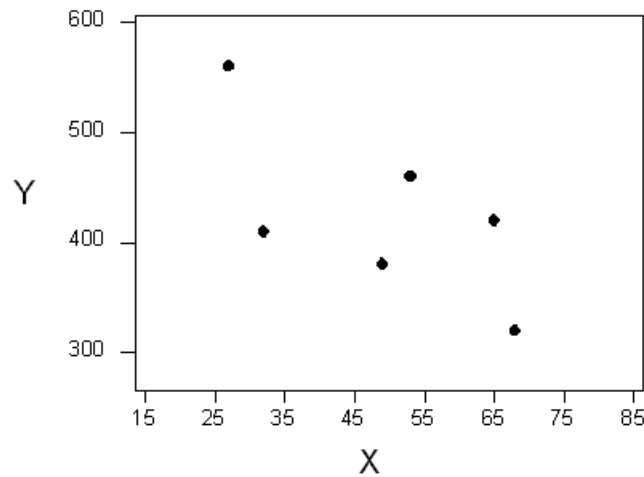
5.5.17 Module 5 | Linear Relationships (6 of 8)

Learning Objectives

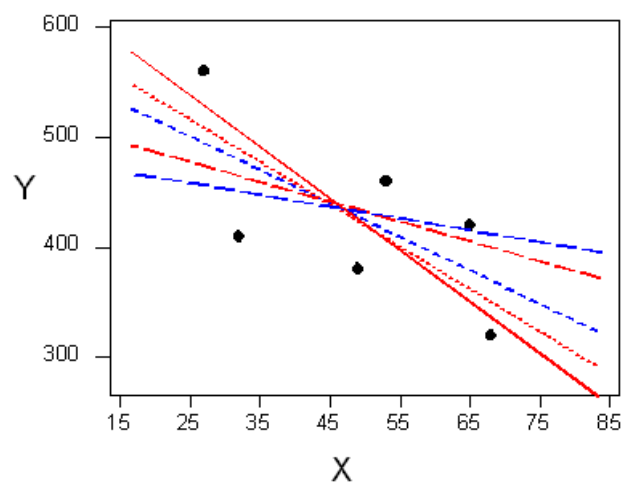
- In the special case of linear relationship, use the least squares regression line as a summary of the overall pattern, and use it to make predictions.

The technique that specifies the dependence of the response variable on the explanatory variable is called *regression*. When that dependence is linear (which is the case in our examples in this section), the technique is called *linear regression*. Linear regression is therefore the technique of finding the line that best fits the pattern of the linear relationship (or in other words, the line that best describes how the response variable linearly depends on the explanatory variable).

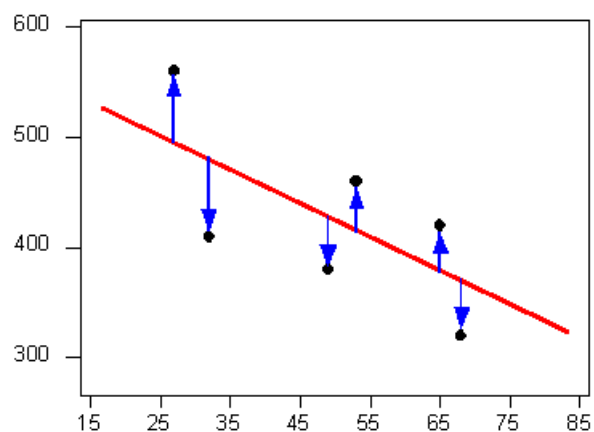
To understand how such a line is chosen, consider the following very simplified version of the age-distance example (we left just 6 of the drivers on the scatterplot):



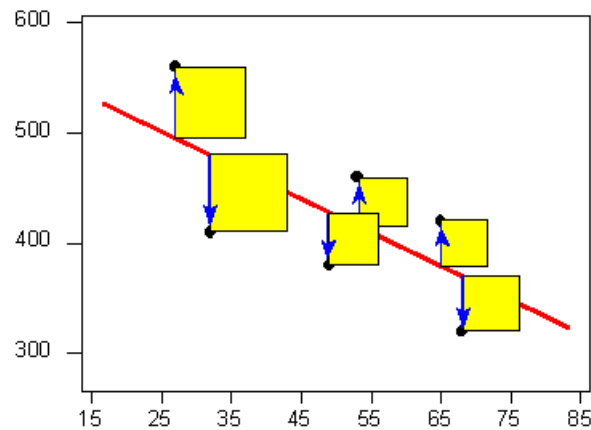
There are many lines that look like they would be good candidates to be the line that best fits the data:



It is doubtful that everyone would select the same line in the plot above. We need to agree on what we mean by “best fits the data”; in other words, we need to agree on a criterion by which we would select this line. We want the line we choose to be close to the data points. In other words, whatever criterion we choose, it had better somehow take into account the vertical deviations of the data points from the line, which are marked with blue arrows in the plot below:



The most commonly used criterion is called the *least squares* criterion. This criterion says: Among all the lines that look good on your data, choose the one that has the smallest sum of squared vertical deviations. Visually, each squared deviation is represented by the area of one of the squares in the plot below. Therefore, we are looking for the line that will have the smallest total yellow area.



This line is called the *least-squares regression line*, and, as we'll see, it fits the linear pattern of the data very well.

For the remainder of this lesson, you'll need to feel comfortable with the algebra of a straight line. In particular you'll need to be familiar with the *slope* and the *intercept* in the equation of a line, and their interpretation.

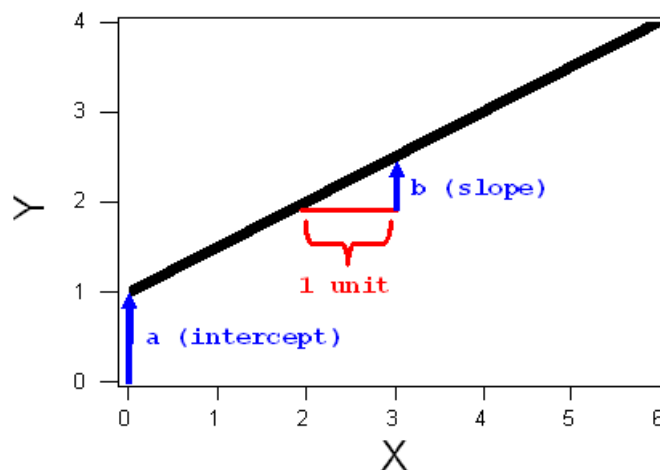
5.5.18 Algebra Review

The algebra of a line A line is described by a set of points (X, Y) that obey a particular relationship between X and Y . That relationship is called the equation of the line, which we will express in the following form: $Y = a + bX$

In this equation, a and b are constants that can be either negative or positive. The reason to write the line in this form is that the constants a and b tell us what the line looks like, as follows:

1. The *intercept* (a) is the value that Y takes when $X = 0$.
2. The *slope* (b) is the change in Y for every increase of 1 unit in X .

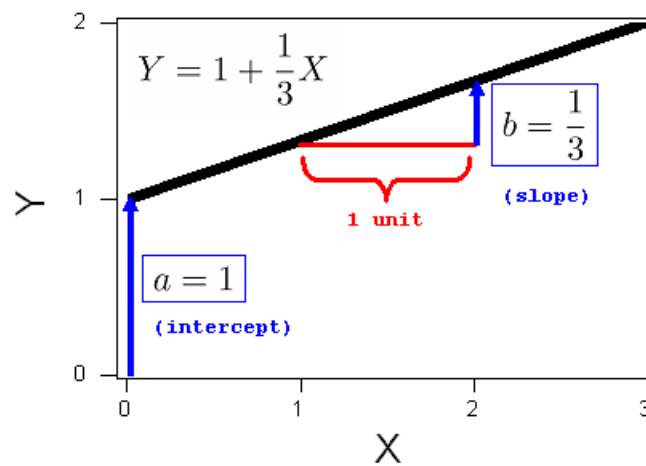
The slope and intercept are indicated with arrows on the following diagram:



Note that if either the intercept (a) or the slope (b) is negative, the corresponding blue arrow on the diagram above would point downward.

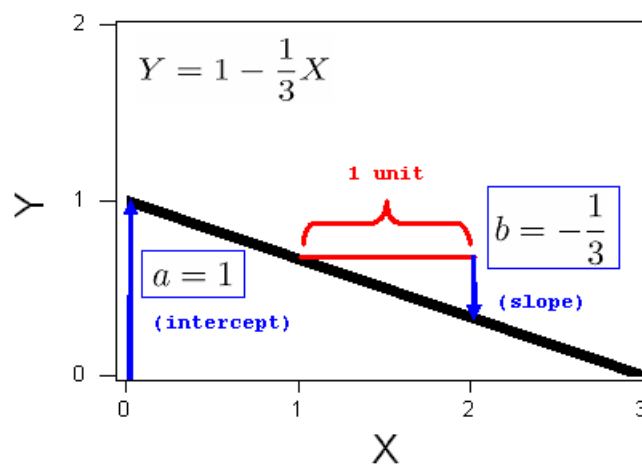
Example (H)

Consider the line $Y = 1 + \frac{1}{3}X$. The intercept is 1. The slope is $1/3$, and the graph of this line is, therefore:



Example (I)

Consider the line $Y = 1 - \frac{1}{3}X$. The intercept is 1. The slope is $-1/3$, and the graph of this line is, therefore:



5.5.19 Module 5 | Linear Relationships (7 of 8)

Learning Objectives

- In the special case of linear relationship, use the least squares regression line as a summary of the overall pattern, and use it to make predictions.

Like any other line, the equation of the least-squares regression line for summarizing the linear relationship between the response variable(Y) and the explanatory variable (X) has the form: $Y = a + bX$ All we need to do is calculate the intercept a , and the slope b , which is easily done if we know:

- $\bar{X}$ – the mean of the explanatory variable's values
- S_X – the standard deviation of the explanatory variable's values
- $\bar{Y}$ – the mean of the response variable's values
- S_Y – the standard deviation of the response variable's values
- r – the correlation coefficient

Given the five quantities above, the slope and intercept of the least squares regression line are found using the following formulas:

- $b = r \left(\frac{S_Y}{S_X} \right)$
- $a = \bar{Y} - b\bar{X}$

Comments:

1. Note that since the formula for the intercept a depends on the value of the slope, b , you need to find b first.
2. The slope of the least squares regression line can be interpreted as the average change in the response variable when the explanatory variable increases by 1 unit.

🔗 Example (J) Age-Distance

Let's revisit our age-distance example, and find the *least-squares regression line*. The following output will be helpful in getting the 5 values we need:

Statistic	Age (X)	Distance (Y)
Mean	51	423
StDev	21.78	82.8
Correlation	-0.793	

- The *slope* of the line is: $b = -0.793 \frac{82.8}{21.78} = -3$. This means that for every 1-unit increase of the explanatory variable, there is, on average, a 3-unit decrease in the response variable. The interpretation *in context* of the slope being -3 is, therefore: For every year a driver gets older, the maximum distance at which he/she can read a sign decreases, *on average*, by 3 feet.
- The *intercept* of the line is: $a = 423 - (-3 \times 51) = 576$ and therefore the *least squares regression line* for this example is:

$$\text{Distance} = 576 + (-3 \times \text{Age})$$

Here is the regression line plotted on the scatterplot:



As we can see, the regression line fits the linear pattern of the data quite well.

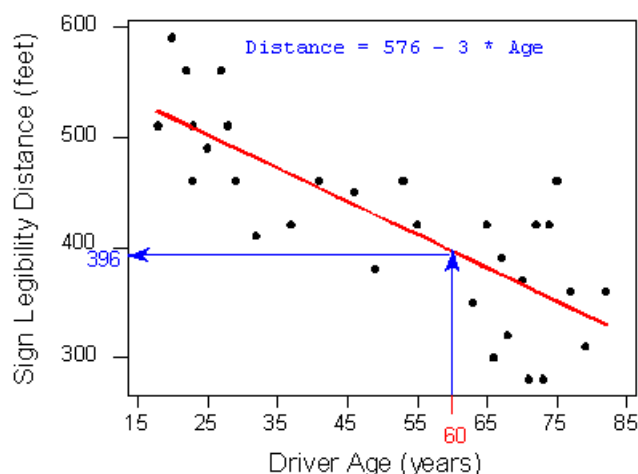
Comment As we mentioned before, hand-calculation is not the focus of this course. We wanted you to see one example in which the least squares regression line is calculated by hand, but in general we'll let a statistics package do that for us.

5.5.20 Module 5 | Linear Relationships (8 of 8)

Learning Objectives

- In the special case of linear relationship, use the least squares regression line as a summary of the overall pattern, and use it to make predictions.

Let's go back now to our motivating example, in which we wanted to predict the maximum distance at which a sign is legible for a 60-year-old. Now that we have found the least squares regression line, this prediction becomes quite easy:

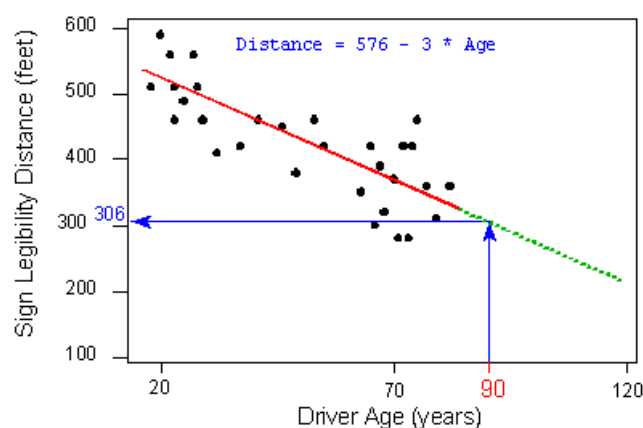


Practically, what the figure tells us is that in order to find the predicted legibility distance for a 60-year-old, we plug $\text{Age} = 60$ into the regression line equation, to find that:

$$\text{Predicted distance} = 576 + (-3 \times 60) = 396$$

396 feet is our best prediction for the maximum distance at which a sign is legible for a 60-year-old.

Comment About Predictions Suppose a government agency wanted to design a sign appropriate for an even wider range of drivers than were present in the original study. They want to predict the maximum distance at which the sign would be legible for a 90-year-old. Using the least squares regression line again as our summary of the linear dependence of the distances upon the drivers' ages, the agency predicts that 90-year-old drivers can see the sign at no more than $576 + (-3 \times 90) = 306$ feet:



(The green segment of the line is the region of ages beyond 82, the age of the oldest individual in the study.)

Question: Is our prediction for 90-year-old drivers reliable?

Answer: Our original age data ranged from 18 (youngest driver) to 82 (oldest driver), and our regression line is therefore a summary of the linear relationship *in that age range only*. When we plug the value 90 into the regression line equation, we are assuming that the same linear relationship extends beyond the range of our age data (18-82) into the green segment. *There is no justification for such an assumption.* It might be the case that the vision of drivers older than 82 falls off more rapidly than it does for younger drivers. (i.e., the slope changes from -3 to something more negative). Our prediction for $\text{age} = 90$ is therefore *not reliable*.

In General Prediction for ranges of the explanatory variable that are not in the data is called *extrapolation*. Since there is no way of knowing whether a relationship holds beyond the range of the explanatory variable in the data, extrapolation is not reliable, and should be avoided. In our example, like most others, extrapolation can lead to very poor or illogical predictions.

5.5.21 Let's Summarize

- A special case of the relationship between two quantitative variables is the *linear* relationship. In this case, a straight line simply and adequately summarizes the relationship.
- When the scatterplot displays a linear relationship, we supplement it with the *correlation coefficient* (r), which measures the *strength* and direction of a linear relationship between two quantitative variables. The correlation ranges between -1 and 1 . Values near -1 indicate a strong negative linear relationship, values near 0 indicate a weak linear relationship, and values near 1 indicate a strong positive linear relationship.
- The correlation is only an appropriate numerical measure for linear relationships, and is sensitive to outliers. Therefore, the correlation should only be used as a supplement to a scatterplot (after we look at the data).
- The most commonly used criterion for finding a line that summarizes the pattern of a linear relationship is “least squares.” The *least squares regression line* has the smallest sum of squared vertical deviations of the data points from the line.
- The slope of the least squares regression line can be interpreted as the average change in the response variable when the explanatory variable increases by 1 unit.
- The least squares regression line predicts the value of the response variable for a given value of the explanatory variable. *Extrapolation* is prediction of values of the explanatory variable that fall outside the range of the data. Since there is no way of knowing whether a relationship holds beyond the range of the explanatory variable in the data, extrapolation is not reliable, and should be avoided.

5.6 Module 5 | Causation

5.6.1 Module 5 | Causation and Lurking Variables (1 of 5)

Learning Objectives

- Recognize the distinction between association and causation, and identify potential lurking variables for explaining an observed relationship.

Introduction So far we have discussed different ways in which data can be used to explore the relationship (or association) between two variables. To frame our discussion we followed the role-type classification table:

		Response	
		Categorical	Quantitative
Explanatory	Categorical	✓ C→C	✓ C→Q
	Quantitative	✗ Q→C	✓ Q→Q

and we have now completed learning how to explore the relationship in cases $C \rightarrow Q$, $C \rightarrow C$, and $Q \rightarrow Q$. (As noted before, case $Q \rightarrow C$ will not be discussed in this course.) When we explore the relationship between two variables, there is often a temptation to conclude from the observed relationship that changes in the explanatory variable *cause* changes in the response variable. In other words, you might be tempted to interpret the observed association as causation. The purpose of this part of the course is to convince you that this kind of interpretation is often *wrong!* The motto of this section is one of the most fundamental principles of this course:

Principle Association *does not* imply causation!

5.6.2 Module 5 | Causation and Lurking Variables (2 of 5)

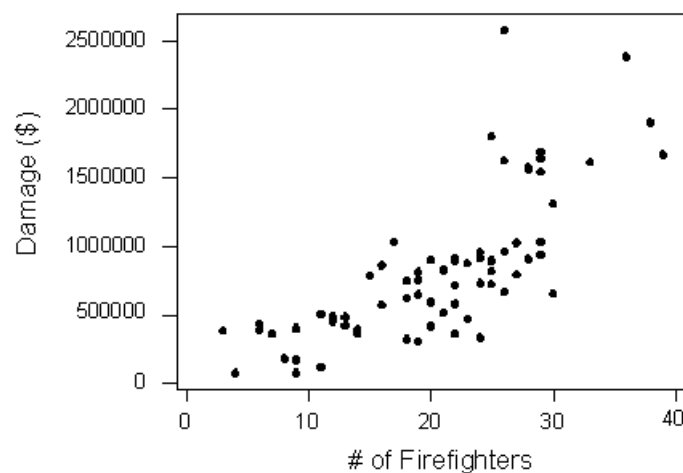
Learning Objectives

- Recognize the distinction between association and causation, and identify potential lurking variables for explaining an observed relationship.

Let's start by looking at the following example:

🔗 Example (K) Fire Damage

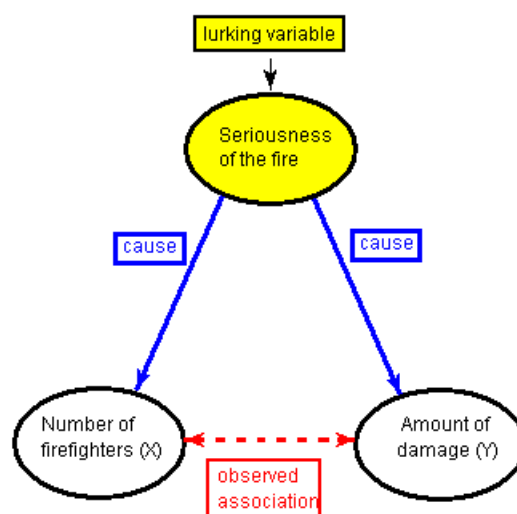
The scatterplot below illustrates how the number of firefighters sent to fires (X) is related to the amount of damage caused by fires (Y) in a certain city.



The scatterplot clearly displays a fairly strong (slightly curved) *positive* relationship between the two variables. Would it, then, be reasonable to conclude that sending more firefighters to a fire causes more damage, or that the city should send fewer firefighters to a fire, in order to decrease the amount of damage done by the fire? Of course not! So what is going on here?

There is a *third variable in the background* – the seriousness of the fire – that is responsible for the observed relationship. More serious fires require more firefighters, and also cause more damage.

The following figure will help you visualize this situation:



Here, the seriousness of the fire is a *lurking variable*. A *lurking variable* is a variable that is not among the explanatory or response variables in a study, but could substantially affect your interpretation of the relationship among those variables.

In particular, as in our example, the lurking variable might have an effect on *both* the explanatory and the response variables. This common effect creates the observed association between the explanatory and response variables, even though there is no causal link between them. This possibility, that there might be a lurking variable (which we might not be thinking about) that is responsible for the observed relationship leads to our principle:

Principle Association *does not* imply causation!

5.6.3 Module 5 | Causation and Lurking Variables (3 of 5)

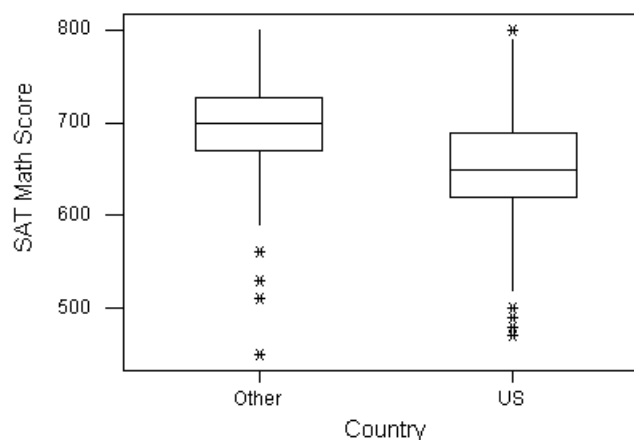
Learning Objectives

- Recognize the distinction between association and causation, and identify potential lurking variables for explaining an observed relationship.

The next example will illustrate another way in which a lurking variable might interfere and prevent us from reaching any causal conclusions.

Example (L) SAT Test

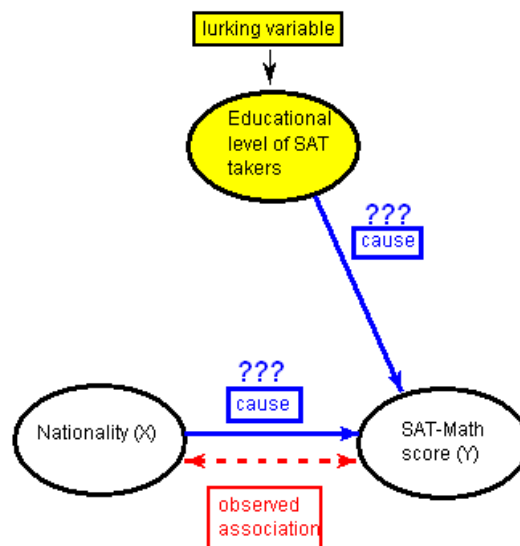
For U.S. colleges and universities, a standard entrance examination is the SAT test. The side-by-side boxplots below provide evidence of a relationship between the student's country of origin (the United States or another country) and the student's SAT Math score.



The distribution of international students' scores is higher than that of U.S. students. The international students' median score (about 700) exceeds the third quartile of U.S. students' scores. Can we conclude that the country of origin is the *cause* of the difference in SAT Math scores, and that students in the United States are weaker at math than students in other countries?

No, not necessarily. While it *might* be true that U.S. students differ in math ability from other students – i.e. due to differences in educational systems – we can't conclude that a student's country of origin is the cause of the disparity. One important *lurking variable* that might explain the observed relationship is the educational level of the two populations taking the SAT Math test. In the United States, the SAT is a standard test, and therefore a broad cross-section of all U.S. students (in terms of educational level) take this test. Among all international students, on the other hand, only those who plan on coming to the U.S. to study, which is usually a more selected subgroup, take the test.

The following figure will help you visualize this explanation:



Here, the explanatory variable (X) *may* have a causal relationship with the response variable (Y), but the lurking variable might be a contributing factor as well, which makes it very hard to isolate the effect of the explanatory variable and prove that it has a causal link with the response variable. In this case, we say that the lurking variable is *confounded* with the explanatory variable, since their effects on the response variable cannot be distinguished from each other.

Note that in each of the above two examples, the lurking variable interacts differently with the variables studied. In example 1, the lurking variable has an effect on both the explanatory and the response variables, creating the illusion that there is a causal link between them. In example two, the lurking variable is confounded with the explanatory variable, making it hard to assess the isolated effect of the explanatory variable on the response variable.

The distinction between these two types of interactions is not as important as the fact that in either case, the observed association can be at least partially explained by the lurking variable. The most important message from these two examples is therefore: *An observed association between two variables is not enough evidence that there is a causal relationship between them.*

In other words ...

Principle Association *does not* imply causation!

5.6.4 Module 5 | Causation and Lurking Variables (4 of 5)

Learning Objectives

- Recognize the distinction between association and causation, and identify potential lurking variables for explaining an observed relationship.
- Recognize and explain the phenomenon of Simpson's Paradox as it relates to interpreting the relationship between two variables.

So far, we have:

- discussed what lurking variables are,
- demonstrated different ways in which the lurking variables can interact with the two studied variables, and
- understood that the existence of a possible lurking variable is the main reason why we say that association does not imply causation.

As you recall, a lurking variable, by definition, is a variable that was not included in the study, but could have a substantial effect on our understanding of the relationship between the two studied variables.

What if we *did* include a lurking variable in our study? What kind of effect could that have on our understanding of the relationship? These are the questions we are going to discuss next.

Let's start with an example:

Example (M) Hospital Death Rates

Background: A government study collected data on the death rates in nearly 6,000 hospitals in the United States. These results were then challenged by researchers, who said that the federal analyses failed to take into account the variation among hospitals in the severity of patients' illnesses when they were hospitalized. As a result, said the researchers, some hospitals were treated unfairly in the findings, which named hospitals with higher-than-expected death rates. What the researchers meant is that when the federal government explored the relationship between the two variables – hospital and death rate – *it also should have included in the study (or taken into account) the lurking variable – severity of illness.*

We will use a simplified version of this study to illustrate the researchers' claim, and see what the possible effect could be of including a lurking variable in a study. (Reference: Moore and McCabe (2003). *Introduction to the Practice of Statistics.*)

Consider the following two-way table, which summarizes the data about the status of patients who were admitted to two hospitals in a certain city (Hospital A and Hospital B). Note that since the purpose of the study is to examine whether there is a “hospital effect” on patients' status, “Hospital” is the explanatory variable, and “Patient's Status” is the response variable.

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	63	2037	2100
	Hospital B	16	784	800
	Total	79	2821	2900

When we supplement the two-way table with the conditional percents within each hospital:

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	3%	97%	100%
	Hospital B	2%	98%	100%

we find that Hospital A has a higher death rate (3%) than Hospital B (2%). Should we jump to the conclusion that a sick patient admitted to Hospital A is 50% more likely to die than if he/she were admitted to Hospital B? *Not so fast ...*

Maybe Hospital A gets most of the severe cases, and that explains why it has a higher death rate. In order to explore this, we need to *include (or account for) the lurking variable “severity of illness” in our analysis.* To do this, we go back to the two-way table and split it up to look separately at patients who are severely ill, and patients who are not.

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	63	2037	2100
	Hospital B	16	784	800
	Total	79	2821	2900

Accounting for the lurking variable:
"severity of illness"

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	57	1443	1500
	Hospital B	8	192	200
	Total	65	1635	1700

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	6	594	600
	Hospital B	8	592	600
	Total	14	1186	1200

As we can see, Hospital A *did* admit many more severely ill patients than Hospital B (1,500 vs. 200). In fact, from the way the totals were split, we see that in Hospital A, severely ill patients were a much higher proportion of the patients – 1,500 out of a total of 2,100 patients. In contrast, only 200 out of 800 patients at Hospital B were severely ill. To better see the effect of including the lurking variable, we need to supplement each of the two new two-way tables with its conditional percentages:

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	3.8%	96.2%	100%
	Hospital B	4.0%	96.0%	100%

		Patient's Status		
		Died	Survived	Total
Hospital	Hospital A	1.0%	99.0%	100%
	Hospital B	1.3%	98.7%	100%

Note that despite our earlier finding that overall Hospital A has a higher death rate (3% vs. 2%), when we take into account the lurking variable, we find that actually it is Hospital B that has the higher death rate both among the severely ill patients (4% vs. 3.8%) and among the not severely ill patients (1.3% vs. 1%). *Thus, we see that adding a lurking variable can change the direction of an association.*

Whenever including a lurking variable causes us to rethink the direction of an association, this is called *Simpson's paradox*.

The possibility that a lurking variable can have such a dramatic effect is another reason we must adhere to the principle:

Principle Association *does not* imply causation!

5.6.5 Module 5 | Causation and Lurking Variables (5 of 5)

Learning Objectives

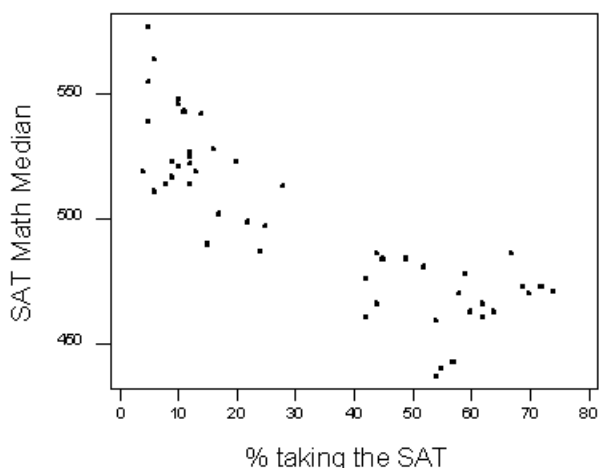
- Recognize the distinction between association and causation, and identify potential lurking variables for explaining an observed relationship.
- Recognize and explain the phenomenon of Simpson's Paradox as it relates to interpreting the relationship between two variables.

It is *not* always the case that including a lurking variable makes us rethink the direction of the association. In the next example we will see how including a lurking variable just helps us gain a deeper understanding of the observed relationship.

Example (N) College Entrance Exams: Including a Lurking Variable

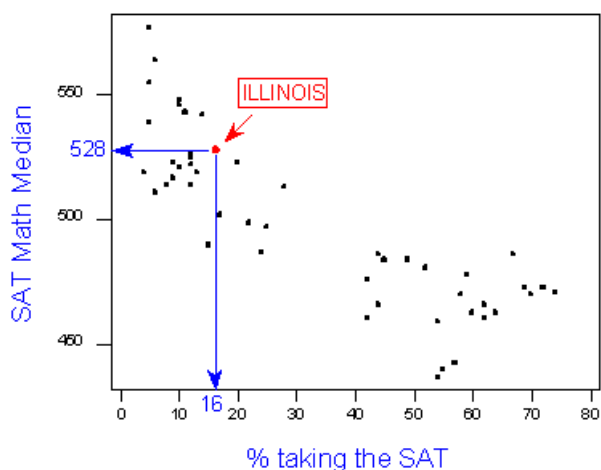
As discussed earlier, in the United States, the SAT is a widely used college entrance examination, required by the most prestigious schools. In some states, a different college entrance examination is prevalent, the ACT.

The following scatterplot displays the relationship between the percentage of students taking the SAT and the median SAT Math scores in each of the 50 U.S. states.

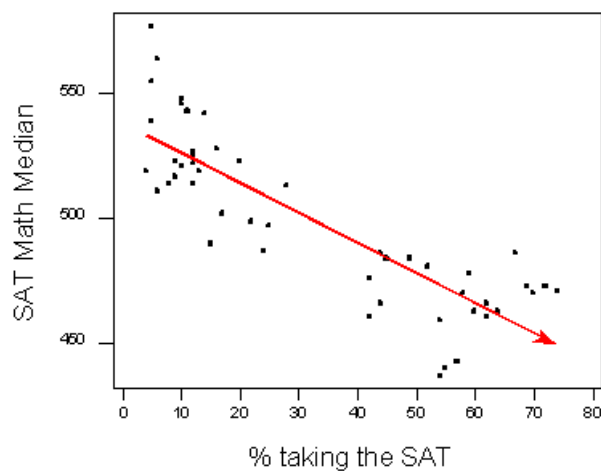


Note that:

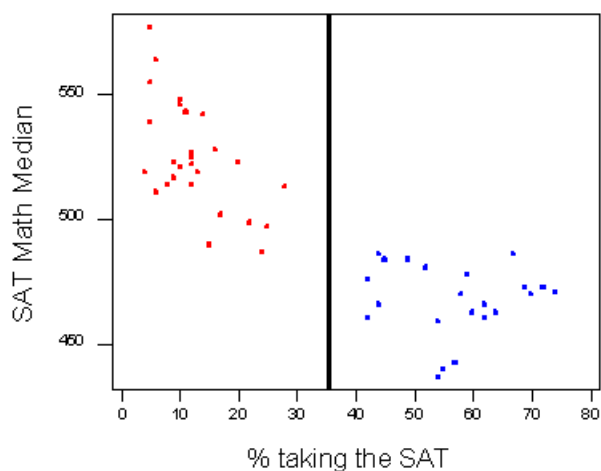
- the explanatory variable is the percentage taking the SAT,
- the response variable is the median SAT Math score, and
- each data point on the scatterplot represents one of the states, so for example, in Illinois, in the year these data were collected, 16% of the students took the SAT Math, and their median score was 528.



Notice that there is a negative relationship between the percentage of students who take the SAT in a state, and the median SAT Math score in that state. What could the explanation behind this negative trend be? Why might having more people take the test be associated with lower scores?

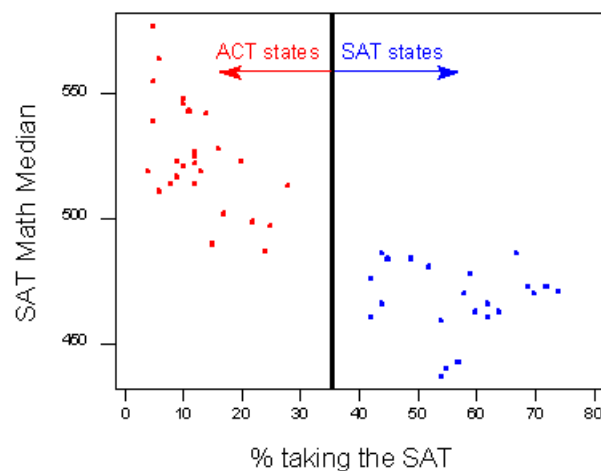


Note that another visible feature of the data is the presence of a gap in the middle of the scatterplot, which creates two distinct clusters in the data. This suggests that maybe there is a lurking variable that separates the states into these two clusters, and that including this lurking variable in the study (as we did, by creating this labeled scatterplot) will help us understand the negative trend.



It turns out that indeed, the clusters represent two groups of states:

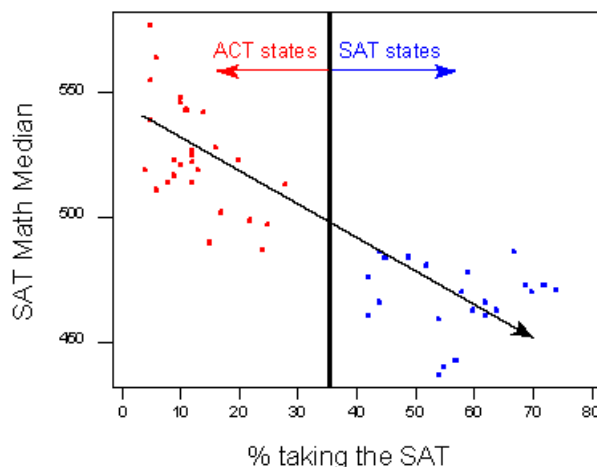
- The “blue group” on the right represents the states where the SAT is the test of choice for students and colleges.
- The “red group” on the left represents the states where the ACT college entrance examination is commonly used.



It makes sense then, that in the “ACT states” on the left, a smaller percentage of students take the SAT. Moreover, the students who do take the SAT in the ACT states are probably students who are applying to more prestigious national colleges, and therefore represent a more select group of students. This is the reason why we see high SAT Math scores in this group.

On the other hand, in the “SAT states” on the right, larger percentages of students take the test. These students represent a much broader cross-section of the population, and therefore we see lower (more average) SAT Math scores.

To summarize: In this case, including the lurking variable “ACT state” versus “SAT state” helped us better understand the observed negative relationship in our data.



The last two examples showed us that including a lurking variable in our exploration may

- Lead us to rethink the direction of an association (as in the Hospital/Death Rate example).
- Help us to gain a deeper understanding of the relationship between variables (as in the SAT/ACT example).

5.6.6 Let's Summarize

- A *lurking variable* is a variable that was not included in your analysis, but that could substantially change your interpretation of the data if it were included.
- Because of the possibility of lurking variables, we adhere to the principle that *association does not imply causation*.
- Including a lurking variable in our exploration may
 - Help us to gain a deeper understanding of the relationship between variables.
 - Lead us to rethink the direction of an association.
- Whenever including a lurking variable causes us to rethink the direction of an association, this is an instance of *Simpson's paradox*.

5.6.7 Summary (EDA)

This summary provides a quick recap of the material you've learned in the Exploratory Data Analysis unit. Please note that this summary *does not provide complete coverage* of the material, but just lists the main points. We therefore recommend that you use this summary only as a checklist or a review before going on to the next unit, or before an exam.

- The purpose of exploratory data analysis (EDA) is to convert the available *data* from their raw form to an informative one, in which the main features of the data are illuminated.
- When performing EDA, we should always:
 - use *visual displays* (graphs or tables) plus *numerical summaries*.

- describe the *overall pattern* and mention any *striking deviations* from that pattern.
- *interpret* the results we got *in context*.
- When examining the *distribution* of a single variable, we distinguish between a *categorical* variable and a *quantitative* variable.
- The distribution of a *categorical* variable is summarized using:
 - Display: pie-chart or bar-chart (variation: pictogram → can be misleading – beware!)
 - Numerical summaries: category (group) percentages.
- The distribution of a *quantitative* variable is summarized using:
 - Display: histogram (or stemplot, mainly for small data sets). When describing the distribution as displayed by the histogram, we should describe the:
 - * Overall pattern → shape, center, spread.
 - * Deviations from the pattern → outliers.
 - Numerical summaries: descriptive statistics (measure of center plus measure of spread):
 - * If distribution is symmetric with no outliers, use mean and standard deviation.
 - * Otherwise, use the five-number summary, in particular, median and IQR (inter-quartile range).
- The five-number summary and the 1.5(IQR) Criterion for detecting outliers are the ingredients we need to build the *boxplot*. Boxplots are most effective when used side-by-side for comparing distributions (see also case C→Q in examining relationships).
- In the special case of a distribution having the normal shape, the Standard Deviation Rule applies. This rule tells us approximately what percent of the observations fall within 1,2, or 3 standard deviations away from the mean. In particular, when a distribution is approximately normal, almost all the observations (99.7%) fall within 3 standard deviations of the mean.
- When examining the relationship between two variables, the first step is to classify the two relevant variables according to their role and type:

		Response	
		Categorical	Quantitative
Explanatory	Categorical	C → C	C → Q
	Quantitative	Q → C	Q → Q

and only then to determine the appropriate tools for summarizing the data. (We don't deal with case Q→C in this course).

- Case C→Q:
Exploring the relationship amounts to *comparing the distributions* of the quantitative response variable for each category of the explanatory variable. To do this, we use:
 - Display: side-by-side boxplots.
 - Numerical summaries: descriptive statistics of the response variable, for each value (category) of the explanatory variable separately.
- Case C→C:
Exploring the relationship amounts to *comparing the distributions* of the categorical response variable, for each category of the explanatory variable. To do this, we use:

- Display: two-way table.
- Numerical summaries: conditional percentages (of the response variable for each value (category) of the explanatory variable separately).
- Case $Q \rightarrow Q$:

We examine the relationship using:

- Display: scatterplot. When describing the relationship as displayed by the scatterplot, be sure to consider:
 - * Overall pattern $\rightarrow$ direction, form, strength.
 - * Deviations from the pattern $\rightarrow$ outliers.

Labeling the scatterplot (including a relevant third categorical variable in our analysis), might add some insight into the nature of the relationship.

In the *special case* that the scatterplot displays a *linear* relationship (and only then), we supplement the scatterplot with:

- Numerical summaries: the correlation coefficient (r) *measures* the direction and, more importantly, the *strength of the linear relationship*. The closer r is to 1 (or -1), the stronger the positive (or negative) linear relationship. r is unitless, influenced by outliers, and should be used only as a supplement to the scatterplot.
- When the relationship is linear (as displayed by the scatterplot, and supported by the correlation r), we can summarize the linear pattern using the regression line. Remember that:
 - * The slope of the regression line tells us the average change in the response variable that results from a 1-unit increase in the explanatory variable.
 - * When using the regression line for predictions, you should beware of extrapolation.
- When examining the relationship between two variables (regardless of the case), any *observed relationship* (association) *does not imply causation*, due to the possible presence of lurking variables.
- When we include a lurking variable in our analysis, we might need to rethink the direction of the relationship $\rightarrow$ *Simpson's paradox*.